International Journal of Greenhouse Gas Control 14 (2013) 183–192

Contents lists available at SciVerse ScienceDirect

International Journal of Greenhouse Gas Control

journal homepage: www.elsevier.com/locate/ijggc

Influence of steam partial pressures in the CO_2 capture capacity of K-doped hydrotalcite-based sorbents for their application to SEWGS processes

Marta Maroño[a,*], Yarima Torreiro[b], Luis Gutierrez[a]

[a] Centre for Energy, Environmental and Technological Research (CIEMAT), Department of Energy, 28040 Madrid, Spain
[b] National Centre for Experimentation in Hydrogen Technologies and Fuel Cells (CNETHPC), 28045 Madrid, Spain

ARTICLE INFO

Article history:
Received 24 July 2012
Received in revised form 9 January 2013
Accepted 10 January 2013
Available online 13 February 2013

Keywords:
K-promoted hydrotalcite-based regenerable sorbents
Pre-combustion CO_2 capture
Sorption enhanced WGS reaction
Influence of steam

ABSTRACT

The performance of three different alkali-promoted hydrotalcite-based materials has been studied under conditions typical of sorption-enhanced WGS process to investigate the influence of steam partial pressure in the feed gas stream in the CO_2 capture mechanisms and system efficiency. The sorbents were prepared by calcination and a minimum temperature of 600 °C was required to guarantee that all CO_2 has been released from the structure. The presence of steam in the feed gas was found to be very beneficial for the capture process efficiency even at very low water content. In order to gain insight into the capture mechanisms involved in the CO_2 capture process, XRD analysis were performed and the results showed that under low steam contents in the feed gas formation of K-dawsonite takes place while under high steam contents (35% v/v) reconstruction of hydrotalcite structure and formation of some carbonates are detected and CO_2 capture values between 4 mol/kg and 9 mol/kg have been measured for the three sorbents when tested under isothermal conditions (300 °C) at $P_{H_2O} = 4.5$ bar and $P_{CO_2} = 0.34$ bar. Under these operating conditions formation of magnesium carbonate is suspected to occur although it could not be demonstrated.

1. Introduction

The development of efficient and low energy consuming processes for CO_2 capture in IGCC processes is an issue of interest nowadays. One of the most promising technologies under investigation is the use of regenerable sorbents to capture CO_2 under the conditions of the water gas shift (WGS) reaction following the so called sorption-enhanced WGS process (SEWGS). This approach combines a high temperature WGS catalyst and a CO_2 sorbent in the same reactor. Adequate sorbents are required to have high CO_2 capture capacity and selectivity towards CO_2 at temperatures in the range of 300–500 °C, adequate sorption–desorption kinetics (easy regenerable), mechanical strength and low cost (Yong and Rodriges, 2002). Hydrotalcites and hydrotalcite-like materials fulfil most of these requisites and they are being studied as suitable candidates for their application under SEWGS conditions (Ding and Alpay, 2001; Van Selow et al., 2009a; Lee et al., 2007a).

Hydrotalcites are compounds that belong to a family of layered double hydroxide solids with the general stoichiometry $M^{2+}{}_{1-x}M^{3+}{}_x(OH)_2A^{m-}{}_{x/m}\cdot yH_2O$ where the divalent ion is typically Mg^{2+}, the trivalent ion is typically Al^{3+}, and the anion (A) is Cl^-, NO_3^- or $CO_3{}^2$. These materials are also known as “anionic clays” and they can accommodate a large variety of di- and trivalent cations and form high surface area and dispersed mixed oxides by thermal decomposition. A very interesting feature is the so-called “memory effect” by which these materials can recover their original structure from the mixed oxides upon contact with water or aqueous solutions containing certain anions. This property provides key aspects for the successful application of hydrotalcite-like compounds as anions exchanges (Pavel et al., 2010) and catalyst support precursors (Rives, 2001). The mechanisms involved in the thermal decomposition and gas-phase reconstruction of Mg–Al hydrotalcites upon contact with water vapor have been studied at room temperature by several authors (Perez-Ramirez et al., 2007; Hibino and Tsunashima, 1998; Erickson et al., 2004; Rocha et al., 1999) but to our knowledge no specific studies of hydrotalcite reconstruction at high temperature can be found in literature.

In general, there is a wide agreement on the benefits of the presence of alkali promoters in the CO_2 capture capacities of hydrotalcite-based materials (Lee et al., 2007b; Oliveira et al., 2008; Ebner et al., 2006) but research is still ongoing to understand the role of the Al^{3+} and K^+ cations content (Walspurger et al., 2008; Meis et al., 2010) or the influence of steam in the CO_2 capture mechanism (Ram Reddy et al., 2008), which is a requisite in the SEWGS process. The concept of SEWGS has already been demonstrated at bench scale by Van Selow et al. (2009b) using potassium promoted hydrotalcites at 400–450 °C and 25 bar under high CO_2 and steam pressure reaching CO_2 capture values as high as 1.6 mol/kg.

* Corresponding author.
E-mail address: marta.marono@ciemat.es (M. Maroño).

1750-5836/$ – see front matter
http://dx.doi.org/10.1016/j.ijggc.2013.01.024

Table 1
Description of materials considered in this study.

Sample	Mg/Al ratio	K_2CO_3 content	Pre-treatment
Puralox MG30 + K_2CO_3	0.5	17 wt%	Two steps heating: 1 h at 250 °C followed by 24 h at 450 °C
Puralox MG61 + K_2CO_3	2	20 wt%	Two steps heating: 1 h at 250 °C followed by 2 h at 400 °C
Puralox MG70 + K_2CO_3HT	3	20 wt%	Hydrated, not calcined

According to these authors those high CO_2 sorption capacity values are associated with the formation of $MgCO_3$ in the bulk of the material. Recently Walspurger et al. (2010) demonstrated the formation of $MgCO_3$ in potassium carbonate promoted hydrotalcites with high magnesium contents (Mg/Al = 2.9) where the presence of high CO_2 and steam pressures and magnesium content in the sample was crucial to guarantee those high CO_2 capture capacities.

In general, reported CO_2 capture capacities of K-promoted hydrotalcites under dry CO_2 gas conditions and/or low steam and CO_2 pressures are lower than those obtained at high CO_2/steam pressures as it is expected for this type of materials where CO_2 capture capacity increased with increasing P_{CO_2} in the feed gas (Allam et al., 2005). However, the mechanisms by which the sorbents capture CO_2 under dry CO_2 gas conditions and/or low steam and CO_2 pressures are still being investigated. Ebner et al. (2006) studied the performance of a K-promoted hydrotalcite (Al/K = 1:1, Mg/Al = 0.75) in the temperature range of 250–500 °C and pressures up to 1.3 bar and obtained a maximum CO_2 working capacity of 0.55 mol/kg. They suggested a CO_2 capture mechanism which included a fast reversible reaction involving the fixation of CO_2 at the material surface followed by other two reversible reactions which involved the capture of CO_2 by forming Mg–K–Al carbonated species. Other authors who agreed with the mechanism proposed by Ebner et al. have reported maximum CO_2 capacities of 0.8 mol/kg obtained at 400 °C and $P_{CO_2} > 0.2$ bar (Lee et al., 2007a) or even as high as 1.4 mol/kg obtained at 400 °C and high pressure (28 bar) for a K-promoted hydrotalcite with Mg/Al ratio = 2.9 (Van Selow et al., 2009a).

Similar CO_2 capture capacities have been reported for potassium promoted hydrotalcite based materials tested in presence of steam although the effect of water and its role in the capture mechanism is still controversial. Thus, Ding and Alpay (2000) found that the presence of water enhanced the CO_2 capture capacity but this effect was independent of the water content and could be seen even at very low water concentrations. To them this fact suggested that the effect of the presence of water was to further activate adsorption sites, possibly maintaining the hydroxyl concentration on the surface or preventing poisoning through carbonate or coke deposition. This positive effect was also observed by Oliveira et al. (2008) who reported a maximum CO_2 capture capacity of 0.75 mol/kg at $P_{CO_2} = 0.4$ and T = 400 °C in presence of at least 16% of water. On the contrary, Ficicilar and Dogu (2006) found that CO_2 capacity of a hydrotalcite material decreased in presence of 20% water due to diffusion resistance of CO_2 to the active sites while Allam et al. (2005) and Hufton et al. (1999) did not see any differences in CO_2 capture capacity of the sorbents in presence of 15–18% of steam in the feed gas.

Under SEWGS conditions the sorbent is combined with a high temperature WGS catalyst and the presence of excess steam is required in order to guarantee the selective conversion of CO to CO_2 and H_2 avoiding the occurrence of non-desired secondary reactions such as methanation or deproportionation reactions (Xue et al., 1996). For high temperature WGS Fe–Cr based catalysts an optimum H_2O/CO ratio of 2 has been widely accepted although values as low as 1.6 have also been used at bench scale without the appearance of secondary reactions (Sánchez et al., 2012). These lower values would allow a reduction of the energetic costs of the process if it is applied at industrial scale. From the point of view of minimising the steam requirements in the SEWGS process a sorbent with a high selectivity towards CO_2 in the presence of water is desired so only the required steam in the WGS reaction will be needed.

From the results described above it seems that there exists a critical CO_2/steam pressure which might be influencing the mechanism of capture of CO_2 on K-promoted hydrotalcites for their application to SEWGS process. At high CO_2 and steam pressures and P_{H_2O}/P_{CO_2} close to 1 the mechanism proposed in literature involves the formation of $MgCO_3$ although recently it has been reported that this is not desired for it causes the decrease of carbon capture rate over time (Van Selow et al., 2011). At low steam pressures the existence of a two-step process is the most generally accepted hypothesis: one fast reversible chemical reaction followed by other slow reversible reaction(s) where different carbonated species are involved but where the role of steam is still not clear.

In this work the thermal behaviour and CO_2 capture capacities of three potassium carbonate promoted hydrotalcite-based materials with different Mg/Al ratio and potassium content are studied. The effect of calcination temperature, process temperature, system pressure and water content in the feed gas are investigated in order to gain insight in the mechanism of CO_2 capture in the temperature range of 300–500 °C and steam pressures in the range of 0–4.55 bar. The best material in terms of CO_2 sorption capacity and selectivity to CO_2 in presence of water has been selected to be used in a SEWGS process.

2. Materials and methods

2.1. Sorbents

Three commercial K_2CO_3-doped hydrotalcite-based materials, supplied by Sasol, have been considered in this study. All samples were provided in pellets of 5 mm × 5 mm and have different Mg/Al ratios (between 0.5 and 3), K_2CO_3 content (17 wt% or 20 wt%) and different heating pre-treatment. Table 1 describes the main characteristics of each material.

2.2. Characterisation of the sorbents

Powder X-ray diffraction patterns were obtained using an X'Pert-MPD PHILIPS powder diffraction apparatus equipped with automatic divergence slit using Cu-Kα radiation. Qualitative analyses were done by comparison with entries from the ICDD (International Centre Diffraction Data) PDF2 2002 database and ICSD Database (Inorganic Crystal Structure Database 2011).

The morphology of samples was determined by Scanning Electron Microscopy (SEM) using an S-2500 HITACHI equipment. The samples were dried overnight at 100 °C and sputter-coated with gold before performing the analysis.

The porous texture characterisation of all materials was assessed by physical gas adsorption, nitrogen and CO_2 adsorption at −196 and 0 °C respectively, in an Autosorb-6 apparatus (by Quantachrome Corp.). The samples were out gassed at 250 °C under vacuum for 4 h. Apparent BET surface area and micropore volume were calculated by fitting nitrogen adsorption data to BET and Dubinin–Radushkevich equations, respectively. Narrow micropore volume (pore size smaller than 0.7 nm)

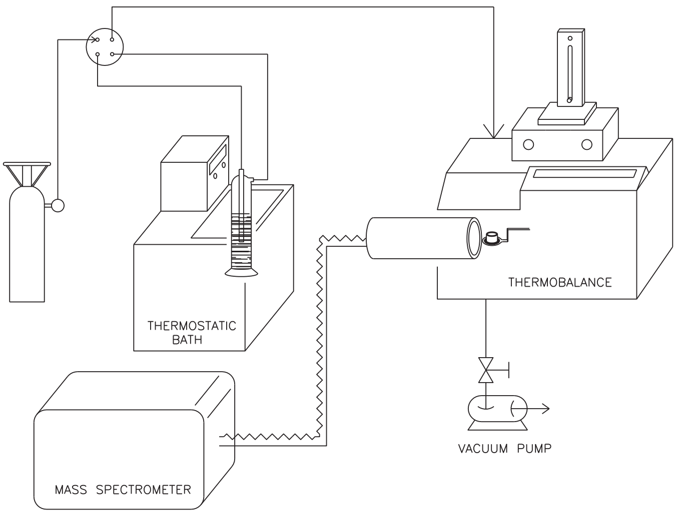

Fig. 1. Flow diagram of the saturation system designed to prepare the wet feed gas used in the screening CO_2 capture test carried out in the thermo balance.

was calculated from CO_2 adsorption at 0 °C using the Dubinin–Radushkevich equation.

2.3. Experimental method

2.3.1. Thermo-gravimetric analysis

Thermo gravimetric analysis of as-supplied samples was performed in a thermobalance METTLER TOLEDO TGA/SDTA 851^e. In each test approximately 50 mg of sample was placed in a platinum basket and heated in N_2 flow (100 ml/min) from room temperature to up to 900–1000 °C at a heating rate of 10 °C/min. Mass spectroscopy was performed with a Thermo Star TM GSD 301T and the results were analysed using the software Quadstar 422.

2.3.2. Preparation of sorbents

Calcination was the procedure used to prepare the sorbents and a calcination programme was designed based on the thermo gravimetric results obtained for all the materials studied. All samples were calcined at temperatures in the range of 500–700 °C in air at chosen temperatures for 4 h and then kept in a dryer overnight.

2.3.3. CO_2 capture screening tests

A first series of screening tests was carried out in the thermobalance to investigate the performance of the different materials in presence of CO_2 in the range of temperatures of interest. In a typical run about 50 mg of sample was placed in a platinum basket, heated in N_2 (100 ml/min) up to 600 °C at a heating rate of 10 °C/min and held at that temperature for 1 h. Then, the heater was shut off and the sample was cooled down to room temperature using a flow of 100 ml/min of CO_2/N_2 (15% CO_2). Fig. 1 shows a flow diagram of the equipment used to perform the screening tests. A saturation system consisting of a thermostatic bath with an impinger was designed to allow the stream of CO_2/N_2 to be saturated in water prior to enter the thermobalance. A two position manual valve allowed the change between dry and wet feed gas depending on the operating mode. Saturation temperature is held at 20 °C so the amount of water fed in each test is around 1.7 mg/min. CO_2 capture tests were carried out under both dry and wet conditions.

2.3.4. CO_2 isothermal capture tests

Isothermal CO_2 capture experiments were carried out for all materials in the temperature range of interest, i.e. 300–500 °C. A computerised Microactivity Pro Unit equipped with a fixed bed reactor was used and a piston pump was coupled to the system to add water to the gas mixture. A detailed description of the system can be found somewhere else (Maroño et al., 2010).

All tests were performed using 10 g of sample and an adsorption temperature of 300 °C and a desorption temperature of 500 °C were used. A gas mixture of 4% CO_2/N_2 and different steam pressures (to up to 4.55 bar of steam) was used. The procedure followed in all tests was the same: first, the sample was out gassed in N_2 at 500 °C for 1 h, then the reactor was cooled down to the adsorption temperature, i.e. 300 °C and pressurised to the desired total pressure (1–15 bar). Adsorption capacity in mol CO_2/kg sorbent was used to evaluate the CO_2 capture capacity of the materials and this value was calculated from the breakthrough curves in the isothermal adsorption tests.

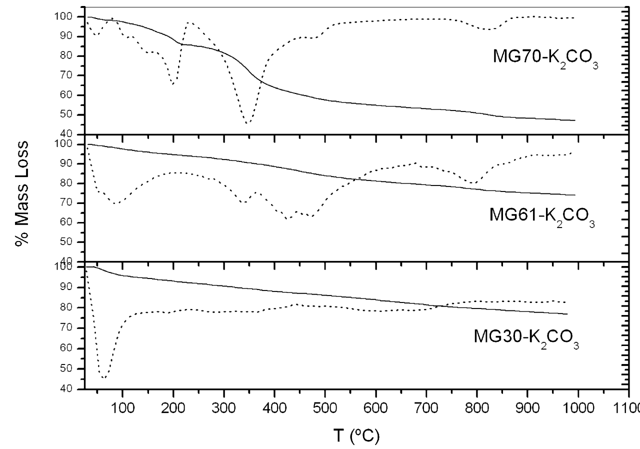

Fig. 2. Thermo-gravimetric analysis profiles for the materials considered in this work.

3. Results

3.1. Thermo gravimetric analysis

Thermal decomposition of hydrotalcite-based materials under atmospheric pressure follows a generally accepted pathway: in a first step, at temperatures below 150 °C, a dehydration process causes the release of superficial and interlayered water. Then, at temperatures between 250 °C and 400 °C, simultaneously or consecutively, dehydroxilation and decarbonation processes take place releasing structural H_2O and CO_2 (Hudson et al., 1995; Roelofs et al., 2002). However, the exact temperatures at which this sequence of events takes place depends on several parameters such as the preparation method and composition (Mg/Al ratio, the presence of promoters) or the thermal treatment profile.

Fig. 2 shows the TG curves obtained for the three hydrotalcite samples under study in this work and Fig. 3 shows the corresponding X ray diffractogram. As expected, differences in calcination history of the materials have conditioned their weight loss profile. In the case of sample MG30-K_2CO_3 (heated at 250 °C for 1 h and further heated at 450 °C for 24 h) an initial weight loss is showed at temperatures below 100 °C and then a continuous and smooth weight loss profile has been obtained. XRD profile for this material (Fig. 3a) showed that it has a very amorphous structure where it was possible to identify the presence of magnesium oxide (MgO) and gamma-alumina (Al_2O_3) (ICDD 98-009-9836) but the predominant phase is the oxide Mg (Al) O, hindering the presence of other species such as carbonates or hydrates that are present in the material and whose slow decomposition might be responsible for the TGA profile observed.

Weight loss profile obtained for sample MG61-K_2CO_3 (see Fig. 2) suggested that CO_2 and H_2O might have been released during TG as

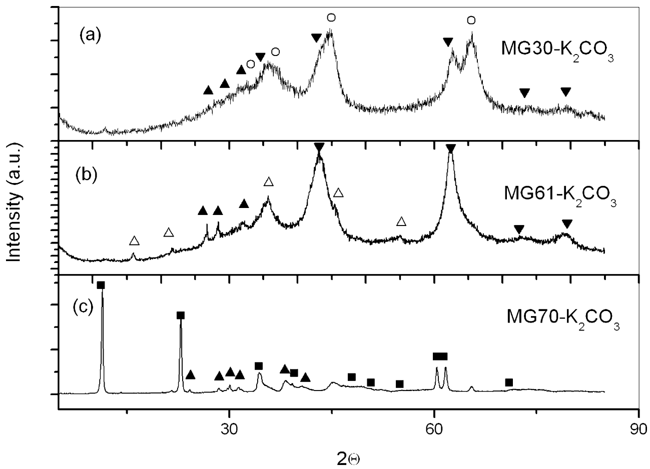

Fig. 3. XRD profiles of as-supplied hydrotalcite samples. (■), hydrotalcite; (△), K-dawsonite; (▼), MgO; (▲), $KH(CO_3)/K_2CO_3 \times 1.5\ H_2O$, (○) Al_2O_3.

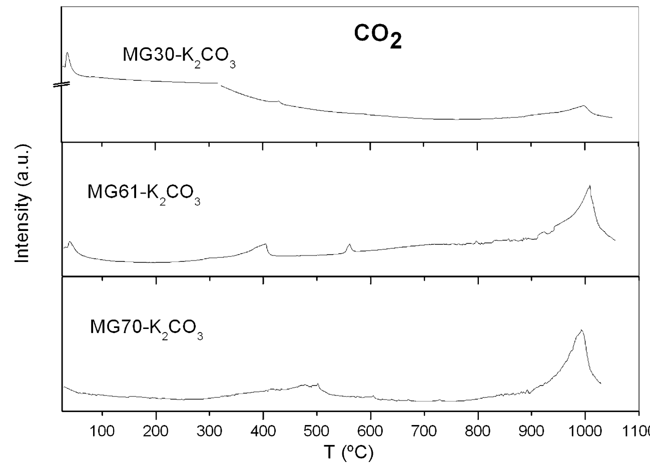

Fig. 4. CO_2 release profile in hydrotalcite samples during TG analysis.

confirmed by the weight loss steps observed at temperatures in the range of 300–600 °C. As this sample has been heated first at 250 °C for 1 h and then at 400 °C for 2 h (according to information provided by the supplier) the dehydroxilation and decarbonation steps might have been still incomplete so additional release of CO_2 and H_2O might be expected under TG conditions. XRD profile (Fig. 3b) shows the formation of MgO, K-dawsonite ($KAl\ (CO_3)_2(OH)_2$) and $K_2CO_3 \times 1.5\ H_2O$. Finally, weight loss profile obtained for material MG70-K_2CO_3 (Fig. 2) approaches that of a typical pattern of the thermal decomposition of a potassium carbonate promoted hydrotalcite-based materials showing an initial weight loss at temperatures below 150 °C and other two between 150 °C and 400 °C which probably correspond to the dedydroxilation and decarbonation processes although in this last case weight loss continues to up to 600 °C. XRD pattern for this sample as supplied corresponds to that of a hydrotalcite material (Fig. 3c).

An additional weight loss at temperatures above 800–900 °C (see Fig. 2) can be observed for all samples which probably correspond to the decomposition of potassium carbonate and other carbonated species.

The weight loss showed by the materials in the temperature range of 300–600 °C during the thermo gravimetric test, suggested that it was due to the release of structural CO_2 and H_2O. In order to confirm this, a mass spectrometer was coupled during the thermo gravimetric analysis of the samples.

The CO_2 release profile for the three samples of hydrotalcite is showed in Fig. 4. At temperatures below 100 °C sample MG30-K_2CO_3 only showed a clear peak of CO_2 which probably corresponds to CO_2 adsorbed in the surface of the material. For the other two samples the release of CO_2 started at temperatures above 300 °C. Sample MG70-K_2CO_3 released CO_2 between 300 °C and 600 °C showing a maximum at 500 °C and reaching a plateau above 600 °C. Sample MG61-K_2CO_3 showed two clear peaks of CO_2, one at 400 °C and another one at 560 °C reaching a plateau above 700 °C. The release of CO_2 above 300 °C can be associated to the loss of anions $CO_2{}^{2-}$ from the decarbonatation of the interlayer structure of the material and to the decomposition of the carbonates of K/Al and Mg/Al formed during the addition of K_2CO_3 and later thermal treatment, being very difficult to distinguish between one and another type (Walspurger et al., 2010). Finally, a clear peak can be observed (see Fig. 4) in the CO_2-MS profile of the three samples at temperatures above 900 °C which corresponds to the release of CO_2 associated to the thermal decomposition of potassium carbonated species present in the samples.

According to the results obtained a pre-calcination temperature above 600 °C will guarantee that the interlayer CO_2 and H_2O have been released from the structure and the materials are ready to be effective in the CO_2 capture process.

3.2. First screening CO_2 capture tests

A first screening CO_2 sorption tests have been carried out for the three hydrotalcite samples pre-calcined at 600 °C under dry and wet CO_2 gas conditions in a thermobalance at atmospheric pressure (Fig. 5).

Fig. 5(a) shows the mass increase obtained for all samples when cooled down from 600 °C to room temperature using a feed gas mixture of 15%CO_2/N_2. As can be seen, despite their differences in Mg/Al ratio and K_2CO_3 content, the curve profile is similar in all the cases: a continuous increase in mass as temperature decreases. The initial very strong increase in mass observed for all the materials suggested that a fast chemisorption of CO_2 took place as soon as the sorbents were put in contact with CO_2 for the first time. This behaviour has already been reported by other authors (Ding and Alpay, 2001).

According to the results showed in Fig. 5(a), material MG61-K_2CO_3 precalcined at 600 °C showed the best performance in the whole range of temperatures.

A second series of tests have been performed to study the CO_2 capture process under low water feed gas conditions. Fig. 5(b) showed the mass increase curves obtained for all materials when cooled them down from 600 °C to room temperature in a CO_2 feed gas stream saturated with water at 20 °C and atmospheric pressure (1.7% v/v steam). In comparison with results obtained under dry CO_2 feed gas conditions (Fig. 5a), even in presence of those low water contents a net positive effect can be observed in the mass increase curve for all materials, being this effect more noticeable for material MG30-K_2CO_3. As part of this mass increase could be due to water adsorption a more detailed study of CO_2 capture under isothermal and wet gas conditions has been included later in this paper. Finally, the abrupt increase in mass observed at temperatures below 200 °C when wet feed gas is used is associated to the adsorption of superficial water.

In order to gain insight into the species which may be involved in the CO_2 capture process under dry CO_2 feed gas conditions, XRD analysis have been performed for all the samples before and after the CO_2 capture process. Fig. 6 shows the diffractograms obtained. XRD pattern of material MG30-K_2CO_3 (Fig. 6a, left column) shows a very amorphous structure where magnesium oxide and some carbonates (such as potassium carbonate which is present in the

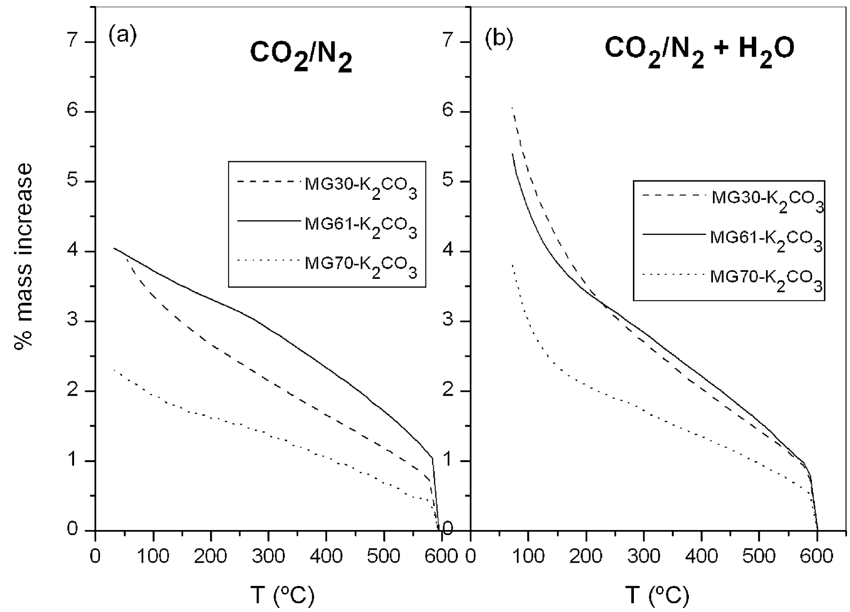

Fig. 5. Mass increase profile obtained for hydrotalcite samples during cooling down from 600 °C to room temperature at atmospheric pressure. Feed gas: (a) 15%CO_2/N_2 and (b) 15%CO_2/N_2 + 1.7% H_2O v/v.

sample) can be hardly identified. XRD patterns of samples MG61-K_2CO_3 and MG70-K_2CO_3 (Fig. 6c and e respectively, left column) are very similar. Both samples have the same K_2CO_3 content (20 wt%). The main difference between them is their Mg/Al ratio which is higher for sample MG70-K_2CO_3 providing a higher crystallinity to the material.

The diffractogram of the samples before the CO_2 capture process showed the presence of hydrated potassium carbonate ($K_2CO_3 \times 1.5$ H_2O) (ICDD no. 11-0655) and K-dawsonite (K $Al(CO_3)_2(OH)_2$) (ICDD no. 21-0979) besides the MgO phase. Our results showed that during thermal decomposition of potassium carbonate promoted hydrotalcites a new phase, the K-dawsonite, was formed and this phase was still present after calcination at 600 °C.

Fig. 7 shows the SEM photographs of the external surface of the three hydrotalcite samples after calcination at 600 °C. Some “agglomerates” in coalescence with some “worm-like” structures can be clearly observed except for material MG30 which amorphous nature may be hiding the presence of carbonated species. Although by EDX was not possible to confirm it, the worm-shaped structures observed in the SEM photographs can be associated to the presence of hydrated potassium carbonate.

After cooling down the three sorbents in a stream of 15%CO_2/N_2 the most remarkable effect was the modification of the

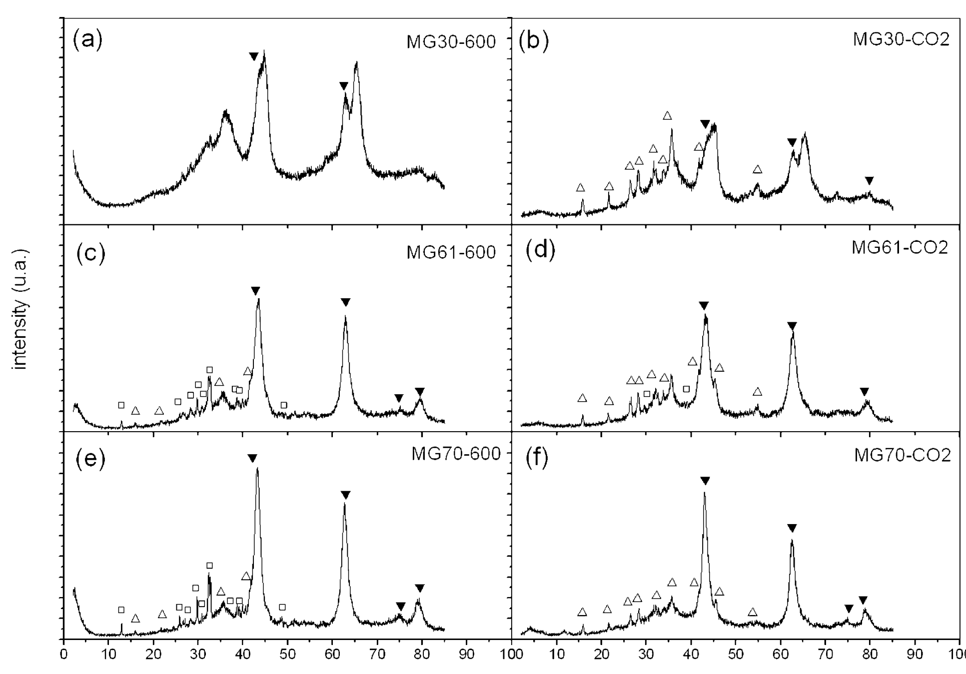

Fig. 6. X ray diffractograms of the three hydrotalcite samples calcined at 600 °C before (left) and after (right) CO_2 capture with CO_2/N_2 in the thermobalance. (□), potassium carbonate; (Δ), K-dawsonite; (▼), MgO.

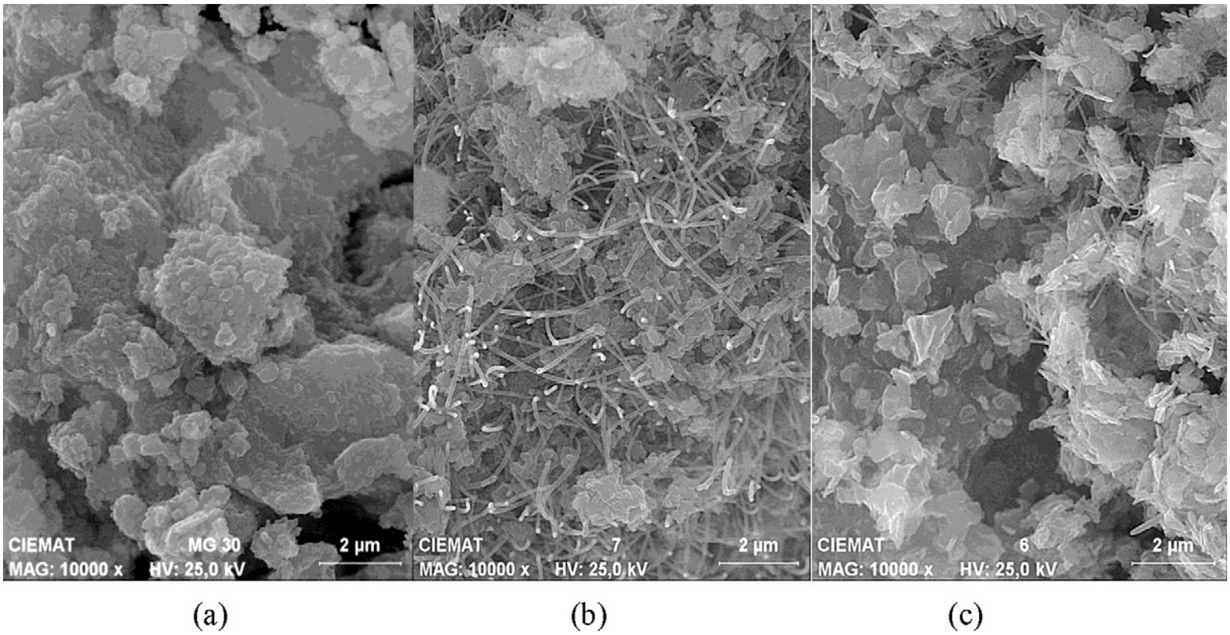

Fig. 7. SEM images of materials after calcination at 600 °C for 4 hours. (a) MG30-K_2CO_3, (b) MG61-K_2CO_3 and (c) MG70-K_2CO_3.

diffractogram in the carbonates peak region (positions 25–40° degrees) (see Fig. 6 right column). The characteristic peaks of K-dawsonite are clearly identified in the three samples while the hydrated carbonate phase is only slightly visible in the MG61-K_2CO_3 sample (Fig. 6d). These results showed that re-formation of K-dawsonite takes place in presence of CO_2. The formation of K-dawsonite has already been observed by Walspurger et al. in alkali-promoted alumina when heated at temperatures up to 300 °C but under relatively high pressure of an equimolar mixture of CO_2 and H_2O (Walspurger et al., 2010). The order followed by the sorbents studied in this work was MG61 > MG30 > MG70 suggesting that values of Al/K ratio approaching the theoretical value of 1 favours the formation of K-dawsonite under the operating conditions tested.

As the presence of hydrated potassium carbonate phase (worm-like structures) in the samples was more clearly observed for material MG61-K_2CO_3, this sample was analysed after the isothermal capture test with dry CO_2 at 300 °C and 1 bar by SEM in order to confirm the conversion of the carbonated phase into K-dawsonite. As showed in Fig. 8 the hydrated potassium carbonate typical needle-shaped structures have completely disappeared. By EDX analysis different phases can be identified: one enriched in Mg and Al which might correspond to the remained structure Mg–(Al)-Ox and another one with a similar content of K and Al which is coherent with the presence of K-dawsonite.

3.3. Isothermal capture of CO_2

According to the CO_2 screening tests performed in the thermobalance for all the materials studied in this work, sorbent MG61-K_2CO_3 seems to fulfil the required characteristics to be used in the proposed system. A series of tests has been done to study the effect of several parameters such as system pressure and water content under isothermal conditions. These tests have been carried out in a fixed bed reactor at 300 °C and 13 bar. An isothermal sorption temperature of 300 °C was chosen as the most adequate to be used in the proposed system. The procedure followed was the same for all the tests and has been described previously in the experimental method section of this paper. In our previous work (Marono et al., 2012) the performance of these three hydrotalcite sorbents was studied under dry feed gas and isothermal conditions and a maximum CO_2 capture capacity of 1.6 mol/kg sorbent was obtained at 300 °C and a $P_{CO_2} = 0.6$ for material MG61-K_2CO_3. In this paper the effect of higher steam contents in the feed gas is explored.

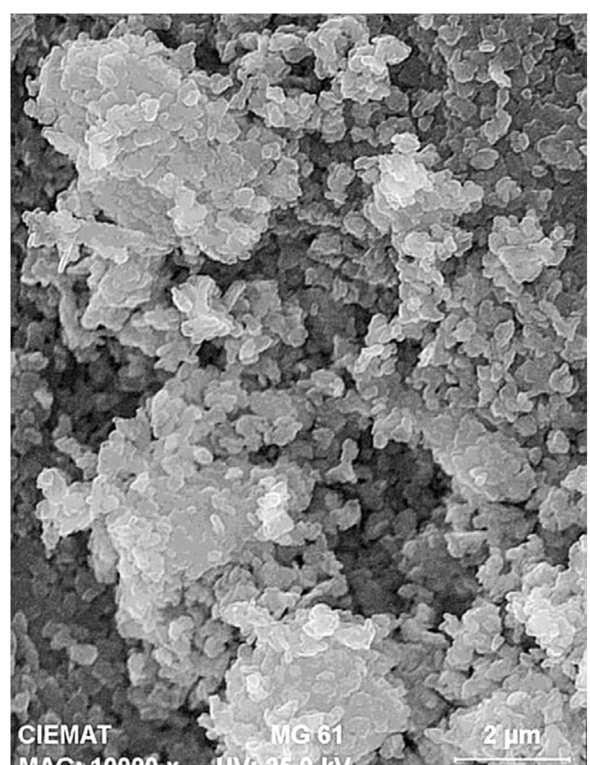

Fig. 8. SEM image of material MG61-K_2CO_3 after isothermal capture of CO_2 at 300 °C and 1 bar.

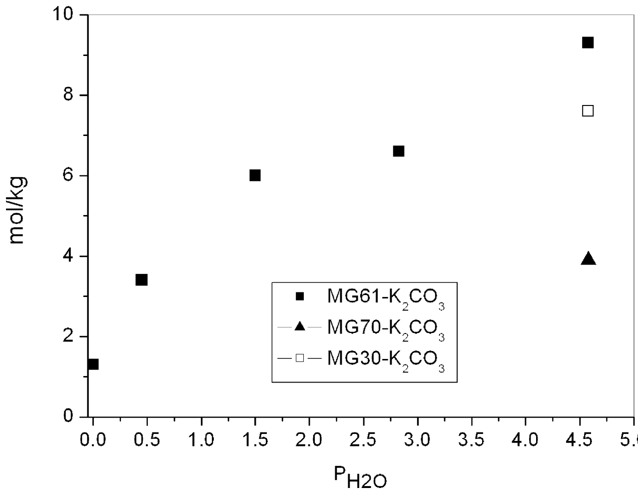

Fig. 9. Effect of partial pressure of steam in CO_2 capture capacity of hydrotalcites. Temperature = 300 °C, $P_{CO_2} = 0.34$ bar.

3.4. *Effect of steam partial pressure*

The presence of high steam contents in the feed gas is a requisite if sorbents are going to be used under SEWGS conditions. In the preliminary tests performed in the thermobalance and showed in Fig. 5(b), a positive effect of the presence of low steam contents in the feed gas was observed for all sorbents. However, the amount of steam available during the WGS reaction is much higher so the behaviour of the sorbents must be studied under those wet conditions. In this section the effect of increasing steam partial pressures in the feed gas to up to 4.5 bar in the capture process was investigated and the results are showed in Fig. 9. According to the results obtained, the CO_2 capture capacity of the sorbents increased with increasing partial pressure of steam reaching values higher than 9 mol/kg at P_{H_2O} of 4.5 bar for the sorbent MG61-K_2CO_3.

As showed in Fig. 9, the performance of the three hydrotalcite samples at $P_{H_2O} = 4.5$ bar and $P_{CO_2} = 0.34$ bar resembles that obtained previously in the thermobalance (Fig. 5b) following the order MG61 > MG30 > MG70 confirming the positive effect of the presence of steam in the CO_2 capture process of these materials. Fig. 10 shows the breakthrough curves profiles obtained for the different hydrotalcite samples when wet feed gas was used. The breakthrough curve for material MG61-K_2CO_3 under dry feed gas has been added for comparison. As it can be observed in Fig. 10, breakthrough time for sorbents MG30-K_2CO_3 and MG61-K_2CO_3 is similar but the shape of the curve is different. In the case of sorbent MG30-K_2CO_3 the curve breaks almost vertically which is typical of chemisorption processes while for sorbents MG61-K_2CO_3 and MG70-K_2CO_3 the breakthrough curves showed a slightly pronounced slope in two steps in the case of material MG61-K_2CO_3 and in one step in the case of material MG70-K_2CO_3.

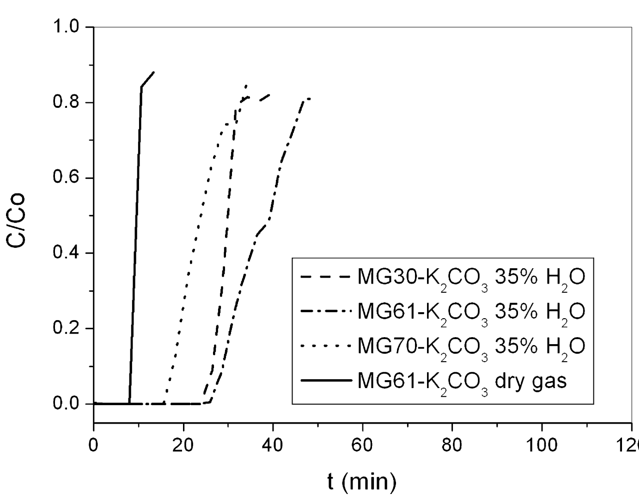

Fig. 10. Breakthrough curves for the three hydrotalcite sorbents in presence of steam. $P_{H_2O} = 4.5$ bar, $P_{CO_2} = 0.34$ bar, temperature = 300 °C, P = 13 bar.

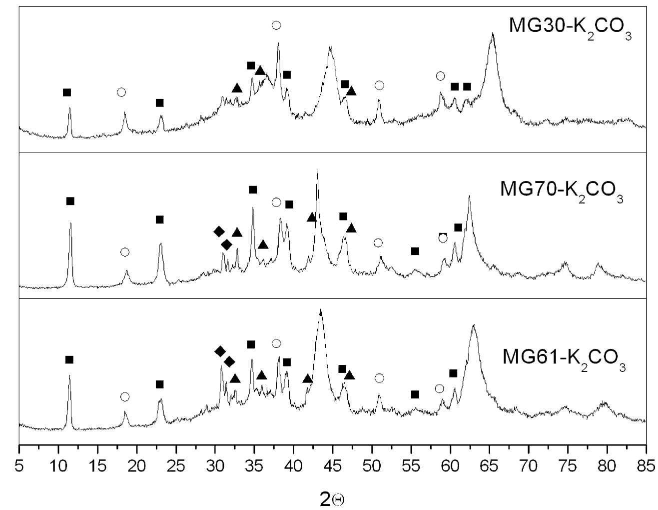

Fig. 11. Reconstruction of hydrotalcite structure in all materials after capturing CO_2 in presence of water. $P_{CO_2} = 0.34$ bar, $P_{H_2O} = 4.5$ bar and carbonates formation (■), Hydrotalcite/Meixnerite; (○), $Mg(OH)_2$ (▲), $K_2Mg(CO_3)_2$; (♦), $K_4H_2(CO_3)_3 \times 1.5\ H_2O$.

In order to explain these differences in behaviour, N_2 adsorption isotherms at 77 K were performed. Specific surface area, pore volume and pore distribution were determined for as-supplied samples and samples calcined at 600 °C. According to the results obtained, all the samples are macro and meso-porous solids describing type II isotherms (data not showed). The S_{BET} values and pore volume obtained followed the order MG30 > MG70 > MG61 while the order obtained for CO_2 sorption capacity (see Fig. 10) was MG61 > MG30 > MG70 so not clear relation could be established between textural properties of the sorbents and their CO_2 capture capacity, in agreement with previous studies reported in literature (Oliveira et al., 2008; Meis et al., 2010).

On the other hand the presence of promoters in the sorbents has been reported to enhance the capture process by chemisorption (Walspurger et al., 2008) so depending on the species formed during calcination and the operating conditions (steam and CO_2 partial pressure) several capture mechanisms might be contributing to the CO_2 capture capacity of the sorbent.

In order to investigate the capture mechanisms involved in the capture capacity of the sorbents under the experimental conditions tested, the three hydrotalcite samples were analysed by XRD after the CO_2 capture experiments in presence of 35% v/v of water. Fig. 11 shows the results obtained. The most remarkable finding is that, as it can be observed in the diffractogram profiles, the characteristic peaks of hydrotalcite, meixnerite and brucite ($Mg(OH)_2$) structures are clearly identified for all the sorbents demonstrating that the "memory effect" concept applicable to this type of materials remains after calcination at 600 °C providing an effective mechanism for CO_2 capture under the SEWGS process conditions. Another relevant result is the modification of the diffractogram profiles in the positions 25–35° which corresponds to the carbonates peak region. As showed in Fig. 11 the main phase identified in this peak zone of the diffractogram was the formation of potassium hydrogen carbonate hydrate K_4 H_2 $(CO_3)_3 \times 1.5$ H_2O (ICDD 20-0886) with a higher peak intensity for sample MG61. Although not clearly distinguished because of overlapping of peaks it was possible to detect the presence of $K_2Mg(CO_3)_2$ (Ref Code ICSD 98-003-1295) which is more clearly seen in sample MG70-K_2CO_3 probably due to its higher crystallinity.

4. Discussion

4.1. Thermal behaviour of sorbents

In the temperature range of interest in this work, between 300 °C and 600 °C, the three materials under study consisted of a mixture of oxides (MgO and Al_2O_3) and carbonates (K-dawsonite and hydrated potassium carbonate). K-dawsonite is a binary carbonate of K and Al which has been reported to be formed in presence of steam and CO_2 during the thermal treatment of alkali promoted alumina at temperatures below 300 °C (Walspurger et al., 2010). Although several authors claim that this compound decomposes a temperatures between 300 °C and 400 °C (Lei et al., 2011) in our work we have observed the presence of K-dawsonite after calcination of the samples at 600 °C as it has been confirmed by XRD patterns (Fig. 6 left). Only in the case of sample MG30-K_2CO_3 the identification of crystalline structures was very difficult due to its amorphous structure so the results obtained by XRD cannot be considered conclusive.

Mass spectrometry profiles of CO_2 obtained for samples MG61-K_2CO_3 and MG70-K_2CO_3 under TGA (Fig. 4) suggested that, in the temperature range of 300–600 °C, the samples followed a thermal decomposition pathway very similar to that proposed by Huggins and Green for dawsonite (Huggins and Green, 1973). According to these authors, dawsonite thermally decomposes in two steps, one very fast between 300 °C and 375 °C where all the hydroxyl water and two thirds of the CO_2 are rapidly given off and carbonates are formed, followed by another step, between 360 °C and 600 °C, where CO_2 is slowly evolved.

Admitting for K-dawsonite a similar thermal decomposition mechanism than for Dawsonite, K-dawsonite will decompose in two step reactions: first, at 300–375 °C the structure collapses releasing very fast all hydroxyl water and approximately 2/3 of the CO_2. Then, the remaining CO_2 is slowly released between 350 °C and 650 °C. TG-MSD results presented before in this paper (Fig. 4) confirmed that in our case the release of CO_2 continues until 600 °C or even higher temperatures. This release of CO_2 at high temperatures has also been observed by Bera et al. (2000) when studying the thermal decomposition of hydrotalcites. This behaviour might explain the increase in the presence of hydrated potassium carbonate observed in our samples as the K-dawsonite phase started to decompose at temperatures higher than 400 °C in the MG70-K_2CO_3 sample and as high as 600 °C in the MG61-K_2CO_3 sample. As no additional water has been used during calcination it seems reasonable to assume that some rearrangement of the hydroxyl groups present in the structure are captured by potassium carbonate during destruction of the original hydrotalcite structure and depending on the alumina content of the sample its persistence is higher (MG70 and MG61 have the same K content). The preservation of CO_3 anions in the hydrotalcite structure at high temperatures has been demonstrated for samples with high Al content (Stanimirova et al., 2004) so the differences in thermal evolution found between samples MG61-K_2CO_3 and MG70-K_2CO_3 can be associated to differences in their Mg/Al ratio.

Thus, in order to prepare adequate sorbents for capturing CO_2 under SEWGS conditions, a minimum calcination temperature of 600 °C is required and depending on the Mg/Al ratio of the material this temperature might be even higher.

4.2. CO_2 capture mechanisms

The CO_2 capture tests performed in this study for three potassium-carbonate promoted hydrotalcite sorbents under dynamic and isothermal conditions have showed that the presence of a different amount of steam available during the CO_2 capture process has a strong influence in the capture process efficiency of the sorbents suggesting that different capture mechanisms might be taking place.

The first series of tests carried out in a thermobalance under dry and very low water content in the feed gas showed that reformation of K-dawsonite took place during cooling down of the samples from 600 °C to room temperature (see Fig. 6). In our previous work (Marono et al., 2012) these sorbents were tested under isothermal conditions and the highest CO_2 capture efficiency obtained at 300 °C under dry feed gas conditions has been 1.6 mol/kg for material MG61-K_2CO_3 pre-calcined at 600 °C. This value is comparable to the highest values published by other authors studying this type of sorbents under similar operating conditions as previously referred in this paper. The breakthrough curve obtained for this sorbent has been included in Fig. 10 for comparison and as it can be seen, it is almost vertical (see Fig. 10) evolving slowly at the end of the sorption process until the feed gas composition was reached again. This behaviour is in agreement with the mechanism proposed by Ebner et al. (2006) who suggested that under dry feed gas conditions the capture mechanism of CO_2 included a fast reversible reaction which fixed CO_2 at the material surface followed by other two reversible reactions which involved the capture of CO_2 by forming Mg–K–Al carbonated species. In our case the fast process might correspond to the interaction between CO_2 and strong basic sites in the mixed oxide produced during the pre-treatment (easily accessible sites) which have been reported to be mainly oxygen anions (Meis et al., 2010; Shen et al., 1994) and the slow process is likely to correspond to the process of intercalation of CO_2 resulting in the re-formation of K-dawsonite.

As no or very low water content was added during the cooling process we must assume that during the slow phase of the capture process reorganisation of the water molecules still present in the material in form of hydrated potassium carbonate occurred at temperatures in the range of 600–300 °C. As no in situ XRD analyses could be performed it was not possible to determine the temperatures at which each process took place but it is reasonable to suppose that as the main phase present in samples pre-calcined at 600 °C was the hydrated potassium carbonate (intermediate phase in the thermal decomposition of K-dawsonite), during the cooling process the CO_2 was slowly being incorporated in the carbonates forming the K-dawsonite again as showed in the diffractogram in Fig. 6 right side.

However, when high steam content in the feed gas is added, the sorbents showed a clear enhancement of the CO_2 capture capacity (see Fig. 10). The three samples of hydrotalcite were tested under isothermal conditions at 300 °C and 13 bar ($P_{CO_2} = 0.34$) and, as showed in our results, when the amount of steam available in the reaction system was increased to up to 4.5 bar the CO_2 capture capacities of the sorbents increased dramatically reaching values as high as 9 mol/kg sorbent for material MG61-K_2CO_3. The analysis by XRD of the samples (Fig. 11) after the capture tests revealed the presence of hydrotalcite structures in the three sorbents confirming that the "memory effect" concept applicable to this type of materials remains after calcination at 600 °C and provides an effective mechanism for CO_2 capture under the SEWGS process conditions. To our knowledge this is the first time that reconstruction of the hydrotalcite structure has been demonstrated at high temperatures (300 °C) for this type of sorbents.

Besides the hydrotalcite structure it was also possible to identify the presence of brucite ($Mg(OH)_2$) and meixnerite

($Mg_6Al_2(OH)_{18} \times 4.5\ H_2O$), a hydrotalcite analogue with OH^- groups as compensating anions in the interlayer instead of the original carbonates which formation has previously been reported for hydrotalcites when reconstructed from the oxides with water vapour or by immersion in decarbonated water (Roelofs et al., 2002; Abelló et al., 2005). The presence of these phases suggested that in our case a defect of CO_2 existed in the reaction media which limited the reconstruction of the structure of hydrotalcite and an optimum ratio P_{CO_2}/P_{H_2O} would be necessary to obtain a complete reconstruction.

The XRD analysis of the samples also showed the presence of potassium hydrogen carbonate hydrate $K_4H_2(CO_3)_3 \times 1.5\ H_2O$ (ICDD 20-0886) with a higher peak intensity for sample MG61-K_2CO_3 providing an additional mechanism for capturing CO_2. The relatively high content in potassium of both samples (20 wt%) and the presence of hydrated potassium carbonate in the sorbents before the capture tests (see Fig. 6 left) justifies the formation of this hydrated phase under the presence of high steam operating conditions. The formation of this phase has previously been reported by Lee et al. (2006) when studying the CO_2 capture capacities of alkali metal-based sorbents in presence of high H_2O concentrations. Although not clearly distinguished because of overlapping of peaks it was possible to detect the presence of $K_2Mg(CO_3)_2$ (ICDD 98-003-1295).

However, the species identified in the XRD patterns in Fig. 11 do not seem to clearly justify the rather high CO_2 capture capacities obtained, especially in the case of those values obtained when high steam partial pressures were used. Moreover, when weighting the samples after the capture tests, the total mass increase measured was in accordance with the amount of water consumed and the mass increase due to the new species formed and detected by XRD but it was lower than the theoretical value which would correspond to the CO_2 capture capacity measured. This suggested that some compounds highly capable to capture CO_2 under high steam partial pressures and CO_2 might be formed during the capture process but which might be desorbed or decomposed during the cooling and/or depressurisation process.

The most plausible explanation for this discrepancy between final sample weight and CO_2 captured is that during the capture process at high steam partial pressures magnesium carbonated species were formed (mostly $MgCO_3$) contributing to the high CO_2 capture capacity observed but they decomposed again during the depressurisation process of the samples (previous to XRD analysis), forming again magnesium oxide. As no in situ XRD analysis could be performed during our experiments we cannot corroborate this hypothesis.

However, during the XRD analysis of our samples after the capture process we identified the presence of MgO and $Mg(OH)_2$ phases (see Fig. 11) which in presence of CO_2 and H_2O under the conditions tested (300 °C and 13 bar) could have formed $Mg(CO_3)$ as reported by Siriwardane and Stevens (2009). Moreover, in a recent work, Walspurger et al. (2010) demonstrated by in situ XRD analysis that under high CO_2 and steam partial pressures magnesium carbonate is formed in potassium carbonate doped hydrotalcite-based materials at 400 °C which completely decomposed when the samples were depressurised to atmospheric pressure in a N_2 stream.

In our experiments, the sorbents were tested at 300 °C and 13 bar and after saturation, the samples were depressurised in N_2 and cooled down before opening the reactor and sending the samples to XRD analysis. Walspurger et al. also reported that relatively high steam partial pressure may induce partial hydroxylation of MgO periclase phase forming magnesium hydroxide (brucite) which in presence of CO_2 seems to react to give magnesium carbonate, being this process more favourable than the dehydration of magnesium hydroxide to periclase (MgO at high working pressures (P = 10 bar)).

Assuming for our materials a similar behaviour of those reported by Walspurger et al. we suggest that $MgCO_3$ could be formed in our experiments, carried out at 300 °C and 13 bar. At high steam partial pressure and under those conditions $Mg(OH)_2$, $MgCO_3$, CO_2 and H_2O coexisted justifying the high CO_2 capture values obtained. When the samples were depressurised and cooled in N_2 decomposition of $MgCO_3$ to MgO took place and only $Mg(OH)_2$ and MgO could be detected by XRD as showed in Fig. 11.

5. Conclusions

The thermal behaviour and CO_2 capture capacities of three K-promoted hydrotalcite-based materials with different Mg/Al ratio and potassium carbonate content are studied. The effect of calcination temperature, process temperature, system pressure and water content in the feed gas has been investigated in order to gain insight in the mechanisms of CO_2 capture in the temperature range of 300–600 °C and steam partial pressures in the range of 0–4.55 bar. Thermo-gravimetric analysis of the samples revealed that a pre-calcination temperature above 500 °C guaranteed that the interlayer CO_2 and H_2O have been released from the structure and the materials were ready to be effective in the CO_2 capture process. Dynamic and isothermal CO_2 capture tests performed under dry and wet feed gas conditions showed that the presence of different amounts of steam during the CO_2 capture process has a strong influence in the capture process efficiency conditioning the capture mechanisms involved. Under dry and low water contents in the feed gas the main capture mechanism identified was the formation of K-dawsonite ($KAl(CO_3)(OH)_2$). However, when the amount of steam available in the reaction system was increased to up to 35% v/v ($P_{H_2O} = 4.5$ bar) the CO_2 capture capacities of the sorbents increased drastically reaching values as high as 9 mol/kg sorbent for material MG61-K_2CO_3. The responsible for these high capture values has been found to be the contribution of several processes: the reconstruction of the hydrotalcite structure which to our knowledge is the first time this mechanism is reported to have been observed in this type of materials at those high temperatures (300 °C), the hydration of potassium carbonate to form the hydrogen carbonate hydrate $K_4H_2(CO_3)_3 \times 1.5\,H_2O$ and the formation of magnesium carbonate during the capture process conditions (300 °C and 13 bar).

According to our results, absolute values of P_{CO_2} and P_{H_2O} and the ratio P_{H_2O}/P_{CO_2} play a crucial role in the CO_2 capture mechanisms and efficiencies of potassium promoted hydrotalcite-based sorbents for their application to SEWGS processes.

Acknowledgements

Authors wish to thank the Ministry of Science and Technology of Spain for his financial support under the CAPHIGAS project (ENE-2009-08002) and the reviewers for their witty comments and suggestions which have contributed to improve the quality of this paper.

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